Volume VII

Advanced Logic

Lambda Calculus

Self-Study Notes

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From Cantor to Ito

Front Matter

Series. From Cantor to Ito

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Chapter 1

Lambda Calculus

lambda-calculus

Type Theory → **Lambda Calculus** → Algebras of Sets

1.1 Lambda Terms

Section Toolkit — Quick Reference

Concept	**One-line summary**
λ-term	Variable x, abstraction $\lambda x.M$, application MN
α-equivalence	$\lambda x.M =_\alpha \lambda y.M[y/x]$ when y fresh
β-reduction	$(\lambda x.M)N \to_\beta M[N/x]$
Normal form	Term with no β-redex; may not exist (non-termination)
Church boolean **T**	$\lambda x.\lambda y.x$
Church boolean **F**	$\lambda x.\lambda y.y$
Church numeral $\overline{n}$	$\lambda f.\lambda x.f^n(x)$ (f applied n times to x)
Type $\tau \to \sigma$	Function type: input of type τ, output of type σ
Typing judgment	$\Gamma \vdash M : \tau$ reads "M has type τ in context Γ"
Curry–Howard	Propositions are types; proofs are terms; $A \to B$ is implication

Untyped Lambda Calculus

Background References

- **First-order logic:** Enderton [1972], Barwise [1977a].
- **General topology:** Kelly [1955].
- **Set theory:** Halmos [1960], van Dalen et al. [1978].
- **Recursion theory:** Rogers [1967].
- **Category theory:** Arbib and Manes [1975], MacLane [1972].

Definition — Lambda Terms

Definition 1.1.1 (Lambda Terms)**.** The set Λ of λ-terms is defined inductively as follows:

1. Every variable x is a λ-term.
2. If M is a λ-term and x is a variable, then
$$\lambda x.\, M$$
is a λ-term.
3. If M and N are λ-terms, then
$$MN$$
is a λ-term.

Remark (Standard quantified statement)*.*

$M \in \Lambda \quad \iff \quad M$ is generated from variables by finitely many applications of $\lambda x.(\,\cdot\,)$ and $(\,\cdot\,)(\,\cdot\,)$.

Remark (Interpretation)*.* The λ-terms are the smallest set closed under three constructions: variables, abstraction (function formation), and application (function evaluation). This inductive definition plays the same role as recursive definitions in analysis or algebra: every λ-term is built from variables by finitely many applications of abstraction and application.

Remark (Structural View)*.* Every λ-term has a tree structure:

- variables are leaves,
- abstractions $\lambda x.\, M$ are unary nodes,
- applications MN are binary nodes.

Thus Λ is a freely generated syntactic algebra.

Proofs

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Lambda Calculus, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Bibliography